Jason (Zheng Jian) Li

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SKILLS

Languages: Python, C/C++, CUDA, SYCL, OpenCL
ML/Inference: PyTorch, vLLM, SGLang, TensorRT-LLM, Triton, FlashInfer
Systems: Linux, SLURM, Docker/Enroot, Git, distributed systems, GPU profiling
Interests: LLM inference, speculative decoding, agents, mechanistic interpretability, AI safety

PROFESSIONAL EXPERIENCE

NVIDIA — Toronto, ON
Systems Software Engineer — *06/2025 – Present*

- Optimized open-source LLM inference frameworks (primarily vLLM) to ensure day-0 model support on NVIDIA hardware, spanning single-GPU to multi-node prompt/decode disaggregated deployments.
- Contributed to ongoing efforts to optimize large-scale WideEP serving of DeepSeek R1/V3 on GB200, helping drive vLLM to 26.2K prefill and 10.1K decode tokens per GPU second — a 3–5× improvement over H200.
- Contributed to NVIDIA's submissions to **InferenceX**, optimizing full-rack disaggregated deployments.
- Optimized vLLM-based inference for NeMo RL, NVIDIA's open-source post-training library for scalable reinforcement learning (GRPO, DPO) of large language models.

CentML (acquired by NVIDIA) — Toronto, ON
ML Systems Software Engineer Co-op — *09/2024 – 12/2024*

- Implemented pipeline parallelization, quantization, and speculative decoding for CServe, an optimized LLM serving engine built on vLLM.
- Integrated Hidet compiler optimizations for quantized model serving, achieving up to 11× speed-up on models such as DeepSeek V3 and R1.

Intel Corporation — Toronto, ON
Compiler Engineer Co-op — *05/2024 – 08/2024*

- Developed regression tests for instruction count in the ESIMD (Explicit SIMD extension of SYCL) LIT testing framework for Intel GPUs.
- Resolved issues related to the new LLVM offloading model in collaboration with the Tools team; contributions visible in the open-source intel/llvm repository.

Huawei Technologies Canada — Markham, ON
Compiler Engineer Co-op — *05/2023 – 08/2023*

- Benchmarked the GPU compiler for OpenCL programs and identified optimization opportunities in the Heterogeneous Compiler Lab.
- Optimized compiler backend to reduce unnecessary NOP instruction insertion.

EDUCATION

University of Waterloo — 2021 – 2025
Bachelor of Mathematics, Computer Science — GPA: 93/100 (3.96/4.0)

RESEARCH EXPERIENCE

University of Waterloo — *09/2023 – 04/2024*
Research Assistant — Prof. Hongyang Zhang, Haochen Sun

- Co-developed GPU-accelerated zero-knowledge proof systems for DNNs and transformer-based LLMs (zkDL, zkLLM), resulting in publications at CCS '24 and IEEE TIFS.
- Explored sub-2-bit quantization methods for standard LLMs.

University of Waterloo — *09/2022 – 04/2023*
Research Assistant — Prof. Alexander Wong, Amy Tai

- Built 3D models of thin food objects using NeRF-based methods for synthetic data generation as part of the NutritionVerse project; published at WiCV @ ICCV '23.

PUBLICATIONS

Conference Proceedings and Journal Publications

zkLLM: Zero Knowledge Proofs for Large Language Models
Haochen Sun, **Jason Li**, Hongyang Zhang — CCS '24

zkDL: Efficient Zero-Knowledge Proofs of Deep Learning Training
Haochen Sun, Tonghe Bai, **Jason Li**, Hongyang Zhang — IEEE TIFS, 12/2023

Workshop Publications

Mechanistic Interpretability of Binary and Ternary Transformers
Jason Li MI Workshop @ ICML '24

NutritionVerse-Thin: An Optimized Strategy for Enabling Improved Rendering of 3D Thin Food Models
Chi-en Amy Tai, **Jason Li**, et al. WiCV @ ICCV '23